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Series hybrid engine control with gear emulation

Abstract

A control system that control speed of an internal combustion engine (ICE) in a series hybrid vehicle includes an accelerator pedal, a vehicle speed sensor and a controller. The accelerator pedal sends a torque request signal. The vehicle speed sensor sends a vehicle speed signal. The controller receives the torque request signal and the vehicle speed signal. The controller is configured to determine, based on the torque request and vehicle speed signals, a plurality of allowed emulated gear ratios; calculate an engine speed for each allowed emulated gear ratio; determine a most efficient engine torque for the calculated engine speeds of each allowed emulated gear ratio; select a desired emulated gear ratio having a most efficient operating point from the plurality of allowed emulated gear ratios; and control the ICE to operate at a corresponding engine speed associated with the desired emulated gear ratio.

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Background/Summary

FIELD

(1) The present application generally relates to plug-in hybrid vehicles and, more particularly, to a control system and method for controlling engine speed in a series hybrid vehicle to emulate a fixed gear transmission in a charge sustaining mode.

BACKGROUND

(2) A series hybrid vehicle includes at least one drive motor used to propel the vehicle. The drive motor(s) are coupled to a high voltage battery system and generator through a common high voltage architecture. The generator converts mechanical power from an internal combustion engine (ICE) into electrical power to charge the high voltage battery or power the drive motor(s). Unlike in a parallel hybrid vehicle with a mechanical fixed gear transmission, in a series hybrid vehicle, the engine speed can be operated independently from the vehicle speed by the generator. This can result in the ICE operating at a single constant speed and load or a select few constant speed and load points that are most efficient for a given driver power request. This can lead to noise vibration and harshness (NVH) concerns when the resulting engine power is significantly higher than the driver power request. Accordingly, while such series hybrid vehicles work for their intended purpose, there is a desire for improvement in the relevant art.

SUMMARY

(3) According to one example aspect of the invention, a control system that control speed of an internal combustion engine (ICE) in a series hybrid vehicle includes an accelerator pedal, a vehicle speed sensor and a controller. The accelerator pedal sends a torque request signal. The vehicle speed sensor sends a vehicle speed signal. The controller receives the torque request signal and the vehicle speed signal. The controller is configured to determine, based on the torque request and vehicle speed signals, a plurality of allowed emulated gear ratios; calculate an engine speed for each allowed emulated gear ratio; determine a most efficient engine torque for the calculated

engine speeds of each allowed emulated gear ratio; select a desired emulated gear ratio having a most efficient operating point from the plurality of allowed emulated gear ratios; control the ICE to operate at a corresponding engine speed associated with the desired emulated gear ratio.

(4) In some implementations, engine speed is equal to a product of vehicle speed and an allowed emulated gear ratio.

(5) In some implementations, the plurality of allowed emulated gear ratios includes at least five allowed emulated gear ratios each having a range of torque for an associated engine speed.

(6) In some implementations, the controller is configured to determine a plurality of allowed emulated gear ratios based on a determination that the ICE is on.

(7) In some implementations, the controller is configured to operate the ICE at a corresponding engine speed proportional to the vehicle speed signal.

(8) In additional aspects, a generator converts mechanical power from the ICE into electrical power to one of charge a high voltage battery and power the at least one electric drive motor.

(9) According to one example aspect of the invention, a method for controlling a speed of an internal combustion engine (ICE) in a series hybrid vehicle is provided. The method includes receiving, at a controller, a torque request signal and a vehicle speed signal; determining, at the controller and based on the torque request and vehicle speed signals, a plurality of allowed emulated gear ratios; calculating, at the controller, an engine speed for each allowed emulated gear ratio; determining, at the controller, a most efficient engine torque for the calculated engine speeds of each allowed emulated gear ratio; selecting, at the controller, a desired emulated gear ratio having a most efficient operating point from the plurality of allowed emulated gear ratios; and controlling, at the controller, the ICE to operate at a corresponding engine speed associated with the desired emulated gear ratio.

(10) In additional features, engine speed is equal to a product of vehicle speed and an allowed emulated gear ratio.

(11) In additional features, the plurality of allowed emulated gear ratios includes at least five allowed emulated gear ratios each having a range of torque for an associated engine speed.

(12) In other examples, the controller determines a plurality of allowed emulated gear ratios based on a determination that the ICE is on.

(13) In additional features, the controller operates the ICE at a corresponding engine speed proportional to the vehicle speed signal.

(14) In additional features, the series hybrid vehicle further includes a generator that converts mechanical power from the ICE into electrical power to one of charge a high voltage battery and power the at least one electric drive motor.

(15) Further areas of applicability of the teachings of the present application will become apparent from the detailed description, claims and the drawings provided hereinafter, wherein like reference numerals refer to like features throughout the several views of the drawings. It should be understood that the detailed description, including disclosed embodiments and drawings referenced therein, are merely exemplary in nature intended for purposes of illustration only and are not intended to limit the scope of the present disclosure, its application or uses. Thus, variations that do not depart from the gist of the present application are intended to be within the scope of the present application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a functional block diagram of a series hybrid electric vehicle that implements a control strategy for controlling engine speed according to various principles of the present application;

(2) FIG. 2 is a plot illustrating charge sustaining and charge depleting modes showing battery state

of charge and engine behavior according to example series hybrid powertrains;

(3) FIG. 3 is a plot illustrating unconstrained series hybrid engine control that results in an engine operating at a constant engine power (constant speed and load) or a select few constant speed and load points that are most efficient for a given driver power request;

(4) FIG. 4 is a plot illustrating a series hybrid engine control using gear emulation according to examples of the present disclosure;

(5) FIG. 5 is a series of plots showing various parameters over time of the series hybrid engine control using gear emulation of FIG. 4; and

(6) FIG. 6 is a logic flow diagram illustrating a method of controlling the series hybrid engine using gear emulation according to examples of the present disclosure.

DESCRIPTION

(7) As mentioned above, in a series hybrid vehicle, the engine speed and load can be operated independently from the vehicle speed by the generator. When a series hybrid vehicle is operating in a charge sustaining mode, the ICE speed and load can operate independently from the vehicle speed and driver power request. This can result in the ICE operating at a single constant speed and load or a select few constant speed and load points that are most efficient for a given driver power request. This can lead to noise vibration and harshness (NVH) concerns when the resulting engine power is significantly higher than the driver power request. Such NVH is most noticeable during low vehicle speed driving where there is less road noise to mask the engine noise. A disconnect sensation can be experienced by the driver between engine sound, driver power request and vehicle speed.

(8) The instant disclosure provides a vehicle control system and control strategy for controlling engine speed in a series hybrid vehicle to emulate a fixed gear transmission in charge sustaining mode. As will be described in detail herein, engine speed is proportional to the current vehicle speed and the selected emulated gear ratio. Multiple emulated gear ratios are available resulting in similar engine operation to a parallel hybrid vehicle with a mechanical fixed gear transmission. This results in the driver perception that the engine speed and resulting noise is connected to the vehicle speed, which is the expected behavior of past drivers of parallel hybrid and conventional powertrain vehicles.

(9) Referring now to FIG. 1, a functional block diagram of an example series hybrid vehicle **100** (also referred to herein as “vehicle **100**”) that incorporates a vehicle control system **102** for controlling engine speed according to the principles of the present application is illustrated. The vehicle **100** includes an electrified powertrain **104** having one or more electric drive units or motors **116** (e.g., electric traction motors) configured to generate and transfer drive torque to a driveline **108** for vehicle propulsion. The driveline **108** can include a gearbox **120** and a drive axle **122** that drives drive wheels **124**.

(10) The drive motors **116** are used to propel the vehicle **100**. The drive motors **116** are coupled to a high voltage battery **130** and a generator **132** through a common high voltage architecture **134**. The generator **132** converts mechanical power from an internal combustion engine (ICE) **140** into electrical power to charge the high voltage battery **130** or power the drive motors **116**. As identified above, in a series hybrid vehicle (unlike in a parallel hybrid vehicle with a mechanical fixed gear transmission), the engine speed and load can be operated independently from the vehicle speed by the generator **132**. In a series hybrid, the ICE **140** does not have a mechanical path to the drive axle **122** (drive wheels). Instead, the only path to produce power at the drive wheels **124** is through the generator **132** to produce electrical power. The electrical power from the generator **132** flows to the battery **130** for charging, and/or to the drive motors **116** for powering the vehicle drive wheels **124**.

(11) The vehicle control system **102** includes a controller **150** that receives signals from sensor inputs **160** such as a drive input device, e.g., an accelerator pedal **162**, for providing a driver input, e.g., a torque request, to the controller **150** and ultimately to the high voltage architecture **134** and ICE **140**. The sensor inputs **160** can also include a vehicle speed sensor **164** that provides a vehicle

speed.

(12) While the vehicle control system **102** is shown as a single controller **150**, it will be appreciated that more controllers and/or modules, such as a supervisory electric vehicle control module, a battery control module, a motor control module and a chassis stability module, can be utilized to control various vehicle components of the series hybrid vehicle **100**. In this regard, various controllers and modules are configured to communicate with each other, utilizing different sensor inputs **160** and calculated parameters for controlling operation of the vehicle control system **102**.

(13) With additional reference now to FIG. 2, a plot **210** illustrating a charge sustaining **214** and a charge depleting mode **216** is shown. A battery state of charge **220** and engine behavior **222** according to example series hybrid powertrains are illustrated. As is known, plug-in hybrid vehicles can operate in a charge depleting mode **216** and a charge sustaining mode **214**. In a charge depleting mode **216**, the energy from the battery **130** is used primarily (or exclusively) to propel the vehicle **100** such that the battery state of charge decreases. In the charge sustaining mode **214**, the fuel energy is used primarily to propel the vehicle **100** so that the battery energy remains approximately constant such that the battery state of charge is maintained. As shown, the ICE **140** starts when energy has been depleted from the battery **130**. Furthermore, the ICE **140** runs at optimal times to minimize fuel consumption while maintaining state of charge in a narrow range.

(14) Turning now to FIG. 3, a plot **250** illustrating unconstrained series hybrid engine control according to prior art examples that results in an engine operating at a constant engine power (constant speed and load) or a select few constant speed and load points that are most efficient for a given driver power request. Such a strategy can result in the engine producing significantly higher power than the driver power request. As can be appreciated, in some operating conditions, such as driving in a subdivision or other area at low speeds, the engine can be commanded to produce significantly higher power than the driver power request. The condition is undesirable as the sound emanating from the ICE can suggest an unreasonably elevated power output for the given situation. The vehicle control system **102** commands a more appropriate engine power (RPM), and thereby perceived engine sound output for a given situation.

(15) FIG. 4 is a plot **310** illustrating a series hybrid engine control using gear emulation according to examples of the present disclosure. The series hybrid engine control using gear emulation results in the engine speed being controlled to be proportional to the vehicle speed and the selected emulated gear ratio, which results in the perception that the ICE **140** is mechanically connected to the drive axle **122** (drive wheels) with a fixed gear transmission having gears **312**.

(16) The plot **310** includes an emulated gear shift map that is used to “shift” between emulated gears so that the engine **140** can operate more efficiently for the given driver power request. In the example shown, the engine **140** is operated at one of the identified speeds **322**, **324**, **326**, **328** and **330**, proportional to vehicle speed. This relationship is more in line with the behavior of a fixed gear transmission where the engine speed is proportional to the wheel speed and current gear. While the gear shift map generally illustrates a bar for each engine speed, it is appreciated that the engine speeds can be alternatively be represented as lines. Further, while an infinite number of gear ratios could be used, the example shown in the plot **310** uses six emulated gear ratios. The series hybrid engine control provides the expected sensation of engine speed increasing with vehicle speed to the driver.

(17) FIG. 5 is a series of plots **350** showing various parameters over time of the series hybrid engine control using the gear emulation of FIG. 4. The series of plots **350** includes a vehicle speed **352**, the emulated gear selected **354**, an engine speed **356**, an engine power **360**, a generator mechanical power **362**. A combined drive motors mechanical power **366**, a battery power **370**, a combined drive motors torque **374** and a high voltage battery state of charge **380**. It will be appreciated that the respective values of the plots **350** may be different within the scope of the present disclosure.

(18) Turning now to FIG. 6, a method **410** of controlling the series hybrid engine using gear

emulation according to examples of the present disclosure will be described. The method **410** starts at **412**. At **416**, control determines whether the engine **140** is on. If the engine **140** is not on, control loops to **416**. If control determines that the engine **140** is on, at **420** control determines an emulated gear ratio allowed based on a current emulated gear, driver torque request, a vehicle speed and the shift map. At **424**, control calculates engine speed for each emulated gear ratio allowed. At **430**, control determines the most efficient engine operating point (torque) for the resulting engine speed of each emulated gear.

(19) At **434**, control selects the emulated gear ratio allowed that has the most efficient engine operating point. At **438**, control operates the ICE **140** at the selected engine speed and torque. Control ends at **440**. The method **410** using the gear emulation series hybrid engine control results in the speed of the ICE **140** being controlled to be proportional to the vehicle speed and the selected emulated gear ratio. The emulated gear shift map is used to “shift” between emulated gears so that the ICE **140** can operate more efficiently for a given driver power request. The ICE operates at the most efficient point (engine torque) of the most efficient emulated gear allowed (which determines the engine speed).

(20) As used herein, the term controller or module refers to an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

(21) It will be understood that the mixing and matching of features, elements, methodologies, systems and/or functions between various examples may be expressly contemplated herein so that one skilled in the art will appreciate from the present teachings that features, elements, systems and/or functions of one example may be incorporated into another example as appropriate, unless described otherwise above. It will also be understood that the description, including disclosed examples and drawings, is merely exemplary in nature intended for purposes of illustration only and is not intended to limit the scope of the present application, its application or uses. Thus, variations that do not depart from the gist of the present application are intended to be within the scope of the present application.

Claims

1. A control system that controls speed of an internal combustion engine (ICE) in a series hybrid vehicle, the control system comprising: an accelerator pedal that sends a torque request signal; a vehicle speed sensor that sends a vehicle speed signal; and a controller that receives the torque request signal and the vehicle speed signal and that is configured to: determine, based on the torque request and vehicle speed signals, a plurality of allowed emulated gear ratios; calculate an engine speed for each allowed emulated gear ratio; determine a most efficient engine torque for the calculated engine speeds of each allowed emulated gear ratio; select a desired emulated gear ratio having a most efficient operating point from the plurality of allowed emulated gear ratios; and control the ICE to operate at a corresponding engine speed associated with the desired emulated gear ratio.
2. The control system of claim 1, wherein engine speed is equal to a product of vehicle speed and an allowed emulated gear ratio.
3. The control system of claim 1, wherein the plurality of allowed emulated gear ratios includes at least two allowed emulated gear ratios each having a range of torque for an associated engine speed.
4. The control system of claim 3, wherein the controller is configured to determine a plurality of allowed emulated gear ratios based on a determination that the ICE is on.
5. The control system of claim 1, wherein the controller is configured to operate the ICE at a corresponding engine speed proportional to the vehicle speed signal.

6. The control system of claim 1, further comprising a generator that converts mechanical power from the ICE into electrical power to one of charge a high voltage battery and power the at least one electric drive motor.

7. A method for controlling a speed of an internal combustion engine (ICE) in a series hybrid vehicle, the method comprising: receiving, at a controller, a torque request signal and a vehicle speed signal; determining, at the controller and based on the torque request and vehicle speed signals, a plurality of allowed emulated gear ratios; calculating, at the controller, an engine speed for each allowed emulated gear ratio; determining, at the controller, a most efficient engine torque for the calculated engine speeds of each allowed emulated gear ratio; selecting, at the controller, a desired emulated gear ratio having a most efficient operating point from the plurality of allowed emulated gear ratios; and controlling, at the controller, the ICE to operate at a corresponding engine speed associated with the desired emulated gear ratio.

8. The method of claim 7, wherein engine speed is equal to a product of vehicle speed and an allowed emulated gear ratio.

9. The method of claim 7, wherein the plurality of allowed emulated gear ratios includes at least two allowed emulated gear ratios each having a range of torque for an associated engine speed.

10. The method of claim 9, wherein the controller determines a plurality of allowed emulated gear ratios based on a determination that the ICE is on.

11. The method of claim 7, wherein the controller operates the ICE at a corresponding engine speed proportional to the vehicle speed signal.

12. The method of claim 7, wherein the series hybrid vehicle further includes a generator that converts mechanical power from the ICE into electrical power to one of charge a high voltage battery and power the at least one electric drive motor.
